EEG-based Emotion Recognition based on DEAP Dataset with Genetic Algorithm Augmented Multi-Layer Perceptron

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***Abstract*—Emotion Recognition plays a pivotal role within the fields of Affective Computing and Brain-Computer Interface. Emotions are the behavioral responses representing the mental state of a person. Recognizing the emotions of a person is indeed crucial for various aspects of human-computer interactions and understanding and responding to one's mental health. Here the method is developed to identify the emotions based on EEG by applying Genetic Algorithm(GA) enhanced Multi-Layer Perceptron(MLP) is a promising path of research that has the ability to contribute to the field of affective computing and improve our understanding of individual emotional responses. To quantify the emotions the Power Spectral Density features with reference to the valence-arousal scale are employed. The architecture of the MLP is optimized using a Genetic Algorithm. The proposed model achieved relatively high accuracy in classifying emotions, distinguishing between high/low valence and high/low arousal emotions, as well as categorizing emotions into four classes based on valence and arousal scales. The planned emotion recognition model has achieved high accuracy in classifying emotions based on valence and arousal dimensions. Achieving accuracy of 91.10% for Valence(low/high) classification and 91.02% for Arousal(low/high) classification is indeed commendable. Furthermore, achieving an accuracy of 83.52% for classifying emotions into four categories based on both valence and arousal (HVLA, HVHA, LVLA, HVHA). It is observed that the stated outcomes are improved compared to current literature in the field of emotion recognition.**

Keywords—EEG, Emotions, Valence, Arousal, Power Spectral Density(PSD), Genetic Algorithm(GA), Multi-Layer Perceptron(MLP)

I. Introduction

EEG-based emotion recognition holds great promise in enhancing human-computer interaction, affective computing applications, and our understanding of human emotions. Ongoing research in this area continues to advance our ability to accurately and non-invasively recognize and respond to human emotions EEG is a non-invasive and versatile technique for measuring brain activity, making it a valuable tool in various scientific, clinical, and practical applications. Its ability to capture real-time brainwave patterns has broad implications for understanding human cognition, emotions, and neurological function. It has been widely used to study brain functioning and cognitive processes, including emotions. Emotions play a crucial role in various aspects of human life, including personal development, communication, and decision-making. Recognizing and understanding emotions can enhance human-computer interactions by enabling a more natural interface [1].

Emotion recognition involves the identification and classification of emotional states based on various signals or data sources. There are several methods for emotion recognition, each utilizing different types of input data and techniques. Here are some of the commonly used methods: Facial expression analysis, Speech analysis, Text analysis, and Physiological signal analysis [2]. The choice of method often depends on the available data, the application context, and the desired level of accuracy. Many modern methods collectively contribute to advancements in emotion analysis, making it more robust, scalable, and applicable to a wide range of real-world scenarios.

Combining information from multiple sources or modalities is a powerful approach to emotion recognition. It not only improves accuracy but also extends the applicability of emotion recognition systems to diverse real-world scenarios, making them more effective and versatile. Fusion techniques involve integrating data from different modalities to boost the performance and robustness of emotion classification. In many cases, combining multiple sources of information, including physiological signals along with visual and vocal data, can lead to more comprehensive and accurate emotion recognition systems. EEG-based Emotion recognition can involve various taxonomies or approaches, depending on how emotions are categorized and classified. Here are a few common taxonomies used in EEG-based emotion recognition: i) Discrete Emotion taxonomy which is also called a basic model. ii) Dimensional taxonomy, which is also called as Valence-Arousal model [3]. The discrete taxonomy classifies emotions into basic categories such as pleasure, sadness, anger, fear, surprise, and disgust. The dimensional taxonomy of emotions is a psychological framework that categorizes emotions based on two primary dimensions: valence and arousal. These dimensions provide a way to describe and understand a wide range of emotional experiences. Valence represents the positive-negative dimension of emotions. Arousal represents the intensity or activation level of an emotion [4]. Electroencephalography (EEG) has been extensively used for recognizing emotions for several reasons, mainly due to its unique capabilities such as High Temporal Resolution, Non-Invasive Nature, Portable and low cost.

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Kulkarni et al. [5] describes how machine learning techniques like SVM can be applied to multimodal data (combining EEG and physiological signals) for emotion recognition. The use of three levels within the valence and arousal dimensions suggests a more refined approach to understanding and classifying emotions, which can have inferences for several applications, including affective computing, mental health assessment, and human-computer interaction. The results indicate that they achieved an accuracy of 57.0% for valence classification and 52.4% for arousal classification using PSD features. It is evident from the literature that a wide range of algorithms which includes a technique that extends traditional kernel methods by allowing the combination of multiple kernels [6], Naïve Bayes Classifier [7], Support vector Machine [8], and Deep Learning Models such as TCN, CNN, and form of RNN-LSTM[9], [10]. It is important to consider the balance between model complexity and available computational resources when designing and training deep learning models.

The objective of the article is to create a system to identify an emotion by utilizing a genetic algorithm (GA) to optimize a multi-layer perceptron neural network(MLP). The Genetic Algorithm explained by the author in the [11] is used to enhance the neural network architecture. In the article [12] the author explains the idea of applying a GA to create an improved MLP is to influence the evolutionary principles of the GA to search for the best architecture and hyperparameters for the MLP. The work by Jenkins [13] explores different variants of the Genetic Algorithm (GA) and their impact on optimization tasks. In particular, they examine four variations of GA: i) Generational ii) Steady-State iii) Steady-Generational iv) Random GA. Also, they experimented with different combinations of crossover and mutation, which are fundamental operations in GA. The choice of GA variant and genetic operations can significantly influence the performance and convergence speed. The work employs the Generational Genetic Algorithms variant in which each generation consists of a new set of individuals. Single-point crossover method is used for creating the offspring. Using a Genetic Algorithm (GA) to optimize the hidden layer neurons of the MLP is a clever approach. Optimizing the number of neurons in the hidden layer can significantly impact the performance of neural networks, and GA offers a way to automatically search for an optimal configuration.

The authors in [14] and [15] used Power Spectral Density (PSD) features to identify emotional states based on Russell's circumplex model. The emotions are classified into 4 classes named as HVLA, HVHA, LVLA, and HVHA based on both valence and arousal dimensions adds granularity to emotion recognition.

The article is well-structured as follows: Section II introduces the dataset that is used to perform the experiment, and Section III focuses on the methodology used to carry out the work. Section IV and V provides a presentation of the experimental results obtained from applying methodology to the dataset. Section VI wraps up the article and summarizes the key findings and conclusions drawn.

II. DATASET

The DEAP dataset is a publicly available multimodal dataset for determining human emotional states [16]. The EEG signals were recorded using a high-resolution and multi-channel headset known as Emotiv Epoc device. The electrical activity of the brain is collected using 32 active electrodes while participants watch the one-minute-long video with a sampling frequency of 512 Hz. The electrodes are placed on the scalp as per the 10-20 international electrode placement shown in Figure 1. After a series of filtering procedures, the data is down-sampled to 128 Hz. To allow a range of 4.0-45.0 Hz, the data is filtered using a bandpass filter. To assess the emotional state of participants at the end of each video, the subjective scale called Self Assessment Manikin (SAM) [17] is used to measure various emotional dimensions, including valence, arousal, dominance, and liking. Work focuses on Valence and Arousal dimensions to understand the emotional responses of participants to the videos, while disregarding the dominance and liking ratings. These ratings are in the range from 1 to 9, which are categorized into two, for both valence and arousal dimensions. The ratings from 1-5 are mapped to low valence/low arousal states and the ratings from 5-9 are mapped to high valence/high arousal states.

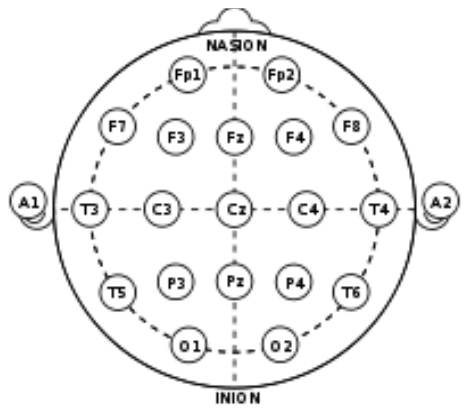

Fig 1. Electrode Positioning System according to 10-20 international system.

In this method, there are three main tasks: EEG Signal preprocessing, feature extraction technique, and classification of emotion which is shown in Figure 2.

A. EEG Signal Preprocessing

Signal processing includes data cleaning, transformation, and normalization. Preprocessing involves removing a baseline of 3 seconds before the actual EEG data related to the experimental task. Then 45 Hz low-pass filter is used to eliminate high-frequency components from EEG data while preserving low-frequency components. Independent Component Analysis(ICA) is used to identify and remove artifacts such as non-brain-related signals that might contaminate the EEG recordings. By isolating and excluding these components, a cleaner EEG signal can be produced. Then Common Average Referencing(CAR) is used to subtract the average voltage across all electrodes from the voltage at each electrode.

B. Feature Extraction Method

Identifying the most operative EEG electrodes for recognizing human emotions is a critical aspect of emotion recognition research. The choice of EEG channels can significantly impact the accuracy and reliability of emotion classification models. It is important to note that the optimal EEG

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channel configuration may vary depending on factors such as the specific emotion being studied, the population under investigation, and the EEG recording setup. The research has indeed shown that different regions of the frontal areas of the brain play significant roles in the processing specifically, the left and right frontal lobes have been associated with distinct aspects of emotional processing. [14], [19]. The EEG channels Fp1, Fp2, F3, and F4 are used in the experiment to realize emotions. These electrodes are typically located in the frontal region of the brain.

Welch's average periodogram method [20] is applied to evaluate the Power Spectral Density (PSD) of signals. In this technique, time-domain signal data is divided into overlapping blocks or sections. Each of these sections is then transformed into the frequency domain using the Fourier transform, resulting in a periodogram for each section. By averaging the periodograms from multiple sections, you get a smoother estimate of the Power Spectral Density (PSD) compared to a single periodogram, which helps to reduce the impact of transient or noisy signals in the data. Brain waves are categorized into different frequency bands based on their oscillation rates. They are Delta waves (0.5-3 Hz), which are slow-frequency oscillations often associated with deep sleep stage. Theta waves (4-7 Hz), connected to drowsiness or daydreaming. Alpha waves(8-13 Hz), are prominent when the brain is in a relaxed but awake state. Beta waves (13-30 Hz) are associated with alertness, concentration, and active thinking. Gamma waves (30-45 Hz) are connected to higher cognitive functions, such as learning, perception, and problem-solving [21].

In the experiment, the average power of the individual frequency band of every channel is considered as a statistical feature. There are four frequency bands for each electrode(Delta band is ignored as it is associated with deep sleep). The mean power of the Theta, Alpha, Beta, and Gamma frequency bands used as a feature. Hence the feature vector is represented as follows:

< Theta_mean of Fp1, Fp2, F3, F4, Alpha mean of Fp1, Fp2, F3,F4, Beta mean of Fp1, Fp2, F3,F4, Gamma mean of Fp1,Fp2, F3, F4 >. So it produces 40(videos)*16(features) .

C. Classification Model

The classification model Multilayer Perceptron (MLP) is employed to assess the connection between EEG data and various emotional states. GA is used to achieve a better neural network architecture as shown in Figure 2. Optimized Multilayer Perceptron (MLP) is used for the classification of emotions into two different schemes: i) Two-class emotions and ii) Four-class emotions.

In GA, for a given problem multiple solutions are figured and are progressed over a number of generations. Every solution consists of a specification that assists in progressing the performance. The weights in a neural network are critical because they determine how the network transforms input data, learns features, and makes predictions to achieve high accuracy.

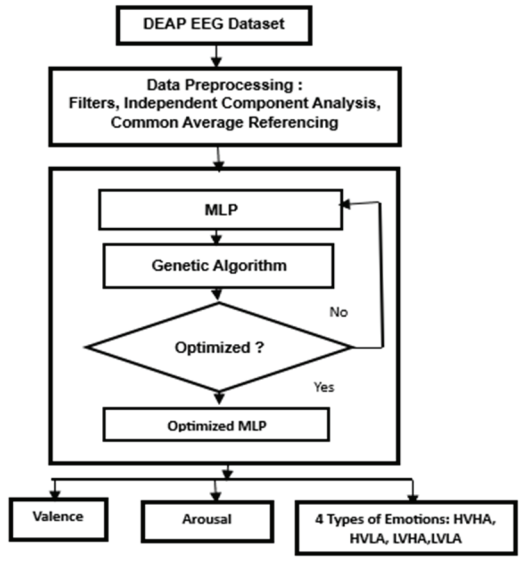

Fig 2. Flowchart for Genetic Algorithm Optimized MLP

The GA applied to the optimization of a neural network, a single solution for a candidate typically consists of all the weights and biases of the neural network. These weights and biases collectively define the architecture and behavior of the neural network, making them the essential parameters to be optimized. The individuals in the population are represented by the total number of nodes of hidden layers. Here in the experiment, the Neural Network architecture has 1,2,1 layers in input, hidden, and output layers respectively. In a Genetic Algorithm (GA), the evolution of generations typically involves three main stages: Selection, Crossover (Recombination), and Mutation. These phases are essential for generating new individuals (solutions) in the population and refining them over successive generations.

1) Stage 1-Selection: The best individuals are selected by the algorithm grounded on the fitness value calculated. To compute the fitness value of candidates in the algorithm, it is required to specify how the MLP is trained, including its hyperparameters and settings. Here are some common parameters you might need to define:

i) To perform the experiment, for hidden layer activation function was selected randomly from Tanh, Sigmoid, Linear and ReLU.

ii) Random selection of Weight optimizers are from the list: Limited - Broyden–Fletcher–Goldfarb–Shanno algorithm [22], SGD [23], and Stochastic Optimizatization [24].

iii) The architecture has two hidden layers, and the each layer size is arbitrarily selected between 2 to 50.

iv) The number of terminations used is 1000. If there is no improvement after 80 iterations, then the classifier is terminated.

In the context of optimization algorithms, including genetic algorithms, the fitness value is a numerical measure that indicates how well a solution (individual) performs on a given task or problem. This fitness value is the accuracy performance metric of the MLP classifier. In the classification model scenario, the GA aims to iteratively improve the accuracy performance of the neural network by selecting and evolving individuals with higher fitness values. This means individuals with better accuracy

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(higher fitness values) are then chosen to be the parents of the next generation.

2) Stage 2-Crossover and Mutation: These are the two elementary operators of the Genetic Algorithm. These operators play a crucial role in introducing genetic diversity and exploring the search space. Through crossover and mutation, the genetic algorithm introduces variations and combinations of parameters, leading to a population of individuals that, over time, is expected to produce neural networks with better and better accuracy. The Crossover is the process of combining the genetic information of two parents to create one or more offspring. Mutation involves making small random changes to one or more parameters of an individual. It is used to raise genetic variability and avoid local minima.

The literature survey stated that tuning the parameters of a genetic algorithm is a critical aspect of solving optimization problems. The parameters that need to be considered are crossover probability, crossover operator, mutation probability, mutation operator, and the number of individuals in each generation.

Normally, there are three methods to implement crossover and mutation [25]. The population size parameter of the genetic algorithm can significantly affect the algorithm's performance. The range of 50 to 100 individuals was suggested by the author in [26]. The choice of mutation and crossover rates is a crucial aspect of tuning a genetic algorithm. The experiment considered a crossover rate of (0.95) which is high and a mutation rate of (0.001) which is low. For each generation, the number of population is 50, as it is suggested by the author[26]. Crossover probability is indeed a value in the range [0, 1]. Running the genetic algorithm for 10 generations to evolve the population and refine the neural network architecture.

III. Results

The results of the proposed work depict there is an improved performance in emotion recognition. This approach can be valuable in various applications, such as human-computer interaction, sentiment analysis, and even in areas like healthcare and virtual reality. Table I describes the classification model's parameters for each participant of the DEAP data set. Various algorithms used to identify the emotions are compared with the proposed in Table II and new/modern methods comparison iin n Table III is done to show the profits of the proposed method. The results of a proposed framework in comparison to traditional machine learning approaches, as presented in Table II, show a significant finding. Realizing an average 10-fold CV with classification metrics accuracy of 91.10% and 91.02%. The use of PSD features in the proposed methodology is a remarkable accomplishment, especially since it has surpassed the performance of existing techniques for 2-class emotion recognition on the DEAP dataset. One known fact is computational cost and processing time a drawbacks of deep learning models in BCI applications. This acknowledgment also sets the stage for discussions on optimizing and adapting the proposed methodology for real-world, computationally constrained scenarios, which can be valuable for practical implementations of BCI systems.

The performance of a classifier in Brain-Computer Interface (BCI) applications indeed relies on several factors like signal quality, feature extraction, artifact removal, algorithm, and training data. Optimizing these factors in a BCI system is a multidisciplinary challenge that involves expertise in signal processing, machine learning, neuroscience, and human-computer interaction. Advances in these areas contribute to the continued improvement of BCI classifier performance.The process of selecting suitable frequency bands and features is a crucial step in the development of effective BCI systems for tasks like emotion recognition. It involves a combination of domain knowledge, experimentation, and optimization to ensure that the methodology can capture and leverage the relevant information in the signals effectively.

IV. Discussion

The results of the proposed work depict there is an improved performance in emotion recognition.. A comparison of the proposed EEG-based Emotion Recognition System with various machine learning techniques (Table II) and modern/advanced methods (Table III) is done to show the profits of the proposed method. The observation that the planned method gained higher classification performance compared to the other methods observed in the literature as shown in Table II, is a significant finding.

In summary, both proper preprocessing through bandpass filtering and careful feature selection are crucial steps in building effective classifiers for time-series data. These steps can significantly impact the classifier's ability to extract relevant information from the data, reduce noise, and ultimately improve classification performance. The choice of appropriate frequency bands and the stability and relevance of selected features are key considerations in enhancing the methodology's performance.

V. Conclusion

The paper introduces a novel approach to automated emotion recognition that involves optimizing a neural network model using a genetic algorithm. The input features for this model are PSD features obtained from channels placed on the frontal brain region. This approach aims to improve the accuracy and effectiveness of emotion recognition systems. It has wide application areas such as psychology, human-computer interaction, and healthcare. The objective of the paper is to maximize the model's performance, measured in terms of classification accuracy, to enhance the accuracy of automated emotion recognition systems. Tested the DEAP dataset with machine learning algorithms and evaluated the performance using accuracy, precision, and recall. The genetic algorithm-optimized MLP with features extracted from the frontal lobe of the brain, outperforms traditional machine-learning algorithms when tested on the DEAP Dataset for emotion recognition.

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Table I. INDIVIDUAL PARTICIPANT CLASSIFICATION ACCURACIES

Subject No.	Valence				Arousal				Four Class			
	Accuracy	**Precision**	**Recall**	**F1**	**Accuracy**	**Precision**	**Recall**	**F1**	**Accuracy**	**precision**	**Recall**	**F1**
S01	87.5	83.33	100	90.90	100	100	100	100	87.5	90.0	87.5	86.11
S02	87.5	100	75	85.71	75	100	60	74.9	75.0	87.5	75	75.83
S03	87.5	100	75	85.71	100	100	100	100	87.5	77.5	87.5	81.94
S04	100	100	100	100	100	100	100	100	100	100	100	100
S05	100	100	100	100	87.5	80	100	88.88	87.5	93.75	87.5	87.5
S06	92.5	92.5	100	95.71	87.5	66.66	100	80	62.5	46.66	62.5	53.33
S07	100	100	100	98.7	100	100	100	100	100	100	100	100
S08	100	100	100	100	87.5	75	100	85.7	87.5	87.5	87.5	87.5
S09	100	100	100	100	87	100	66.67	80	87.5	77.5	87.5	81.94
S10	100	100	100	100	100	100	100	100	100	100	100	100
S11	100	100	100	100	87.5	75	100	85.71	87.5	100	87.5	92.5
S12	87.5	75	100	85.71	100	100	100	100	87.5	90.625	87.5	87.69
S13	100	100	100	100	100	100	100	100	100	100	100	100
S14	87.5	66.66	100	80	87.5	100	100	100	75	66.66	75	70
S15	75	50	50	50	62.5	50	66.66	57.14	50	50	50	50
S16	87.5	100	66.66	80	75	66.66	66.66	66.66	62.5	75	62.5	66.66
S17	75	66.66	66.66	66.66	100	100	100	100	75	75	75	75
S18	100	100	100	100	75	90.83	100	94.57	62.5	45.83	62.5	52.5
S19	97.5	100	95	96.67	100	95	94.17	100	75	56.25	75	64.28
S20	87.5	100	66.66	80	87.5	96	100	98	75	70.83	75	70
S21	87.5	100	80	88.88	87.5	100	50	66.66	87.5	90.62	87.5	86.30
S22	100	100	100	100	100	100	100	100	100	100	100	100
S23	75	100	50	66.7	87.5	100	80	88.88	75	56.25	75	64.28
S24	100	100	100	100	100	100	100	100	100	100	100	100
S25	100	100	100	100	100	100	100	100	100	100	100	100
S26	87.5	100	66.66	80	87.5	80	100	88.88	75	91.66	75	79.10
S27	75	100	50	66.66	100	100	100	100	75	90	75	77.77
S28	87.5	100	75	85.71	75	50	50	50	75	87.5	75	79.16
S29	87.5	100	50	66.67	100	100	100	100	87.5	76.78	87.5	81.73
S30	75	100	50	66.66	100	100	100	100	75	91.66	75	77.08
S31	100	100	100	100	87.5	75	100	85.71	87.5	100	87.5	92.5
S32	87.5	100	75	85.71	100	100	100	100	87.5	91.66	87.5	87.5
Mean	**91.09**	**94.817**	**84.11**	**87.58**	**91.39**	**90.63**	**91.69**	**90.36**	**87.58**	**91.39**	**82.81**	**81.5**

Table II. PERFORMANCE COMPARISON OF THE PROPOSED METHOD WITH DIFFERENT MACHINE LEARNING ALGORITHMS

Classifier Name	Valence			Arousal			Four Types of Emotions		
	Accuracy	**Precision**	**Recall**	**Accuracy**	**Precision**	**Recall**	**Accuracy**	**Precision**	**Recall**
SVM(poly)	60.16	64.40	80.38	66.80	60.27	70.45	41.66	25.31	28.18
SVM (RBF)	56.64	47.86	76.43	66.4	52.73	70.31	40.63	24.87	27.95
XGBoost	64.84	67.71	72.64	68.98	67.1	73.75	44.23	38.85	38.47
Rndom Forest	63.28	63.22	67.47	63.67	58.77	69.97	43.40	38.24	38.02
Linear Regression	67.20	65.06	73.10	61.72	60.62	70.12	42.19	38.26	34.72
KNN	64.45	63.35	70.8	62.89	54.88	73.23	44.28	41.68	35.55
Proposed Method	**91.09**	**94.817**	**84.11**	**91.39**	**90.63**	**91.69**	**87.58**	**91.39**	**82.81**

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Table III. COMPARISON OF THE PROPOSED WITH EXISTING ALONG WITH FEATURES AND CLASSIFIERS

Ref No	Year	Feature	Classifier	Valence	Arousal
[16]	2011	PSD	Naïve Bayes	58	62
[6]	2014	Spectral Power	Multiple-Kernel Learning	66	65
[7]	2015	Spectral Power	Naïve Bayes	62	57
[1]	2016	Biospectrum	Support Vector Machine	61	65
[27]	2017	The first difference of Time series	Support Vector Machine	72	69
[28]	2017	LSTM	LSTM	85	86.
[9]	2021	Spectrogram Representation	Temporal Convolutional Network	69	71
[31]	2020	DWT and FFT	Spiking neural Network	78	74
Proposed	--	PSD	MLP	91.10	91.02

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